Journal of Energy Storage 42 (2021) 103030

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Temperature regulating fibers of high latent heat and strength: Mass production, characterization and applications

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ARTICLE INFO

Keywords:
Temperature regulating fibers
Melting spinning
Latent heat
Mechanical property
Thermostability

ABSTRACT

A kind of temperature regulating fibers (TRFs) with excellent mechanical and thermal properties were prepared by bi-component melt spinning technology on an industrial mass production equipment, in which the core material chips are composed of paraffin wax (PW), polyethylene (PE) and olefin block copolymers (OBC), having the preferable latent heat and thermostability at an onset volatilization temperature of 224.91 °C, and the sheath layer is polypropylene (PP). The prepared TRFs have significant heat absorption and release processes in the range of 20-35 °C with a latent heat up to 26.96 J/g. At the sheath/core ratio of 6:4, the TRFs are 2.09 cN/dtex in the breaking strength and 30.4 % in the elongation after further drawing at a winding speed of 1400 m/min, which fully meet the requirements of the fabric manufacture. After woven with cotton yarn, the TRF/cotton fabrics can effectively reduce the temperature fluctuation by 3 °C in the cooling process, displaying good thermal storage and regulating ability.

1. Introduction

The thermal comfort is not only of great significance to physical and mental health, life safety and production efficiency, but also can effectively reduce the energy consumption of building heating, ventilation and air conditioning system. Phase change materials (PCMs) are a kind of functional materials that absorbs or releases a large amount of latent heat to maintain its temperature at a relatively narrow range by changing the phase state [1–4]. On account of its low cost, high thermal efficiency, reusability and high energy storage, PCMs have the good potential in intelligent temperature regulating applications [5–7].

With the improvement of people's living standards, how to achieve a better thermal comfort of human body has attracted more and more attention. In recent years, the concept of personal thermal management has been proposed, which focuses on the human body itself and its surroundings, and provides high efficiency and low energy consumption solutions for the realization of human thermal comfort [8,9]. Among them, the textiles based on energy storage phase change technology have become hot research topics since 1980s, which can realize the intelligent response to the environmental changes through the active heating/cooling cycles [10–16]. When introducing PCMs into fibers, the temperature regulating fibers (TRFs) and the related textiles can be used to reduce the temperature fluctuations thereby keeping human body comfortable against the surrounding temperature changes.

In current production of the TRFs, compared to the wet spinning process, the melt spinning has short production process, low production cost and environmental friendliness, playing an important role in synthetic fiber industry. However, PCMs usually have the low volatilization temperature and easily leak during the spinning process, which make the preparation of the TRFs by melt spinning become difficult even after years of research. It is noticed that most organic PCMs with the phase change temperature lower than 40 °C usually start to volatilize below 150 °C [17–21], much lower than the conventional melt spinning temperature higher than 220 °C. Therefore, in order to achieve the melt spinning of the TRFs, the current researches focus more on the new spinning devices and composite phase change material of good thermostability.

Many researchers reported the preparation of the TRFs by adding PCM microcapsules into polymers in melt spinning. However the approach requires special microcapsules, and the resulting thermal efficiency is rather low [22]. Iqbal and Sun [23] integrated *n*-octadecane microcapsules into PP fibers by the melt spinning with their latest spinneret. The prepared TRFs contain 8 % of PCM microcapsules and have a latent heat of 8.08 J/g. After a weaving process, the woven

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https://doi.org/10.1016/j.est.2021.103030
Received 26 March 2021; Received in revised form 15 July 2021; Accepted 28 July 2021
Available online 7 August 2021
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fabrics show a significant thermoregulating effect. However, due to the uncontrollable particle size of the microcapsules, it is very difficult to apply PCM microcapsules to TRFs in melt spinning mass production. Moreover, the additive amount of PCM microcapsules was generally limited to less than 10%, which is not enough for temperature regulation performance of textiles.

Recent years, bicomponent melt spinning technology become one of the most interesting developments in the field of man-made functional fibers. Fibrous PCMs can also be prepared by bicomponent melt spinning and used in yarns, nonwovens, woven, and knitted fabrics, and many other application fields [24,25]. Zhang et al. [26] investigated bicomponent fibers with PEG and thickening agents as core, and PP as sheath; they found that a high viscosity of the core part favored stable fiber melt-spinning, and that fast crystallization improved spinnability by decreasing the leakage tendency of the molten core. Another melt-spun bicomponent PCM fiber type was produced with icosane and a polyolefin blend as core, and PP as sheath [27]. The maximum of icosane achieved in the nonwoven was about 21 wt.%, and the resulting fiber achieved a latent heat of 32 J/g.

To achieve TRF filament for woven, Cherif et al. [28] made a great progress in the melt spinning of PP TRFs recently. Owing to the high volatilization temperature of PCM composites, they developed a bi-component melt spinning process with a composite PCM as the core material and PP as the sheath to prepare the TRFs. The obtained PP TRFs exhibit a high latent heat up to 22 J/g at a transition temperature of 44 °C, and its breaking strength was measured to be about 1.35 cN/dtex, which to some extent can be used to weave into fabrics.

The above studies suggest that it is promising to prepare TRFs with the shape-stabilized PCM composites as core in the bi-component melting spinning, however, the difficulty lies in the design and preparation of PCM composites with high thermal stability [29]. In many studies, some polymers and inorganic porous materials such as polyethylene [30,31], polyaniline [32], porous carbon [33,34], graphite [35], graphene or silica aerogel [36,37], were used as substrate for the shape stabilization of PCMs. Among them, polymer-based PCM composites are most suitable for melting spinning because of their good rheological properties [28].

Considering all the properties, PE is a good choice to composite with PCM, Trigui et al. [31] reported that PCM can be stabilized well with PE as substrate. But when using the PW/PE composites as the core material in melt spinning, due to the lower thermostability of the PW/PE composites a mass of PW was volatilized at the spinning temperature above 220 °C [38].

In this study, a novel PCM composites of high thermostability were developed by introducing OBC into PW/PE system, in which paraffin wax (PW) was used as the phase change material with a suitable phase change temperature at 28 °C for human thermal comfort. With the PCM composites as the core and PP as the sheath, the TRFs were prepared by a bi-component melt spinning process and the TRFs/cotton woven fabrics were manufactured in a weaving process on an industrial loom.by the blended TRFs/cotton yarns The related surface morphology, heat storage property, thermostability, mechanical property of the TRFs and TRFs/cotton blended fabrics were detailly investigated. It is confirmed that the prepared TRFs are effective on the temperature regulation and have huge potential in textile production.

2. Experimental

2.1. Chemicals and materials

Polyethylene (PE, melt index 20 g/10min) and polypropylene resin (PP, melt index 25 g/10min) were provided by Sinopec Group (Daqing, China). Olefin block copolymers (DOW OBC INFUSE, 9107) and paraffin wax (PW) were bought from Dow Chemical Company and Laiwu Wenyuan new materials technology Co. Ltd. (Jinan, China), respectively.

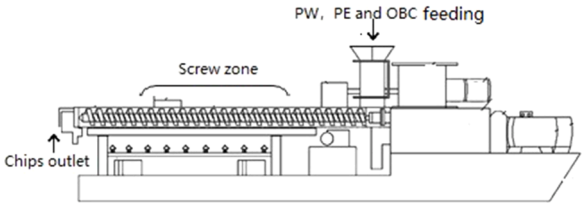


Fig. 1. Schematic diagram of the twin-screw extruder.

2.2. Preparations of the PW/PE-OBC chips and TRFs

Firstly, PW, PE and OBC were mixed uniformly in a certain mass ratio at 150 °C, the PW/PE-OBC chips were then prepared by the melt blending and extruding process on a TSE-30A twin-screw extruder (shown in Fig. 1) at a screw speed of 150 rpm. The required temperatures in each zone of the extruder are shown in Table 1.

After a large number of experiments, combining the latent heat and the rheological properties of the composites, the PW/PE-OBC chips for spinning are prepared at a preferred PW: PE: OBC ratio of 30: 60: 10. Owing to the stable process conditions, the production capacity of the PW/PE-OBC chips reaches 5 kg/h on a machine, which is generally in line with the requirements of mass production.

With the PW/PE-OBC chips as the core and PP as the sheath, the melt spinning process is carried out on a bi-component industrial machine with the sheath-core mass ratios of 7:3, 6:4 and 5:5, respectively. The bi-component spinneret in our study contains 48 capillary holes with a diameter of 0.32 mm. The temperature profiles in the screw zones are set at 210~235 °C for the sheath component and 160~230 °C for the core component. The spinning and the winding speeds are kept constant at 420 m/min and 700 m/min, respectively.

In order to improve the mechanical properties of TRFs, further drawing is implemented at a higher winding speed. In consideration of the mechanical properties and the latent heat value, the TRFs of the sheath-core mass ratios of 6:4 are further drawn at a winding speed of 1400 m/min. After several experiments on the industrial machine, the spinning process could work steadily for a whole day, meeting the requirements of mass production.

2.3. Characterizations

The thermal properties of PW, PW/PE-OBC chips and TRFs were characterized by a differential scanning calorimeter (DSC, Diamond, Perkin Elmer, USA). First, the samples were accurately weighed and sealed into the crucibles, heated to 130 °C at a heating rate of 20 °C/min, held for 3 mins to eliminate the heat history. Then the samples were cooled to 0 °C and subsequently heated up to 50 °C at a rate of 5 °C/min under 40 mL/min nitrogen gas purge.

The thermal decomposition performances of PW, PW/PE-OBC chips and TRFs were analyzed by a thermogravimetric analyzer (TG 209F1 Libra, Netzsch, Germany). All the samples were heated from 30 °C to 550 °C at a heating rate of 10 °C/min with a 20 mL/min nitrogen gas flow.

The cross section and surface morphology of TRFs were observed by a scanning electron microscope (SEM, JSM-5610, JEOL, Japan).

The linear densities of TRFs were measured by a thread length meter (YG086, Wenzhou, China). For each sample, 100 m composite fibers were wound and weighed at room temperature, and then the linear densities were calculated.

According to GB/T 14344-2008 (Chemical fiber filament tensile test method) standard, the mechanical properties of TRFs were tested by a yarn strength meter (XL-2, Shanghai, China). The clamps distance was set to 250 mm under a pre-tension of 0.05 cN/dtex. Each sample was measured for 20 times under a stretching speed of 500 mm/min.

The thermal delay capacity of the TRF/cotton blended woven fabrics

Table 1
The set temperatures in each zone of the twin screw extruder.

Zone	I	II	III	IV	V	VI	VII	VIII	IX	Nozzle
T/(°C)	145	175	185	205	205	205	195	190	170	170
Screw speed	150 rmp									

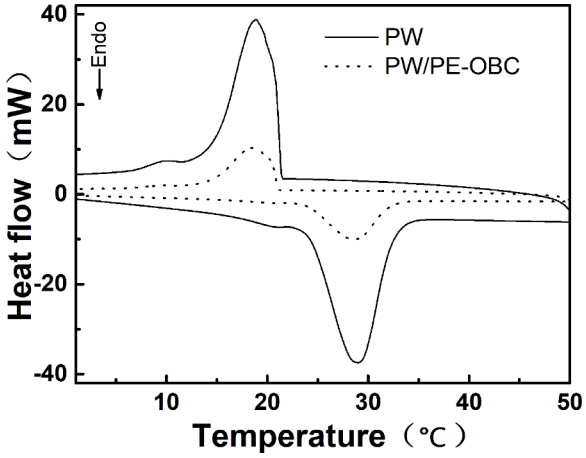


Fig. 2. DSC curves of PW and PW/PE-OBC chips.

Table 2
DSC data of PW and PW/PE-OBC chips.

Sample	T_m (°C)	ΔH_m (J/g)	T_c (°C)	ΔH_c (J/g)
PW	27.26	195.69	23.34	199.13
PW/PE-OBC	28.44	56.22	22.71	57.34

was characterized by a USB temperature data recorder (R4&5, Elitech, USA) with a cooling process from 45 °C to 10 °C.

3. Results and discussion

3.1. Thermal properties of the composite PW/PE-OBC core materials

In melt spinning process, the high thermostability of the PCM composite is of great concern. In view of the melt spinning temperature of PP at about 220 °C, the volatilization temperature of the shape-stabilized PCM must be higher than that to prevent most of PCM escaping from the melt. Therefore, in order to improve the high temperature resistance of PCM, olefin block copolymer (OBC) is introduced in our study.

Based on a lot of experiments, it is found that the shape-stabilized PCM with PW: PE: OBC ratio of 30: 60: 10 can be extruded and granulated smoothly by a twin-screw extruder. The prepared PW/PE-OBC chips not only retain high latent heat, but also have good rheological property for feeding and fiber-forming during the melting spinning.

Fig. 2 shows the DSC curves of paraffin wax (PW) and PW/PE-OBC chips, respectively. The DSC data such as melting peak temperatures (*Tm*), cooling peak temperatures (*Tc*), melting enthalpies (*ΔHm*) and cooling enthalpies (*ΔHc*) are calculated and listed in Table 2.

In Fig. 2, both of PW and PW/PE-OBC chips undergo the melting and crystallization process in the temperature range of 21.34-32.12 °C. Along with the increasing temperature, PW absorbs the latent heat to changes from crystalline state to liquid state. Then, during an exothermic crystallization process between 25.43 °C and 15.82 °C, PW reversely transforms into crystalline state accompanying the release of latent heat. Compared with PW, it is found that the melting and crystallization peaks of PW/PE-OBC chips become significantly weakened due to the only 30 % PW content in the chips.

In Table 2, it is listed that T_m and T_c of PW are 27.26 °C and 23.34 °C with corresponding $\triangle H_m$ and $\triangle H_c$ of 195.69 and 199.13 J/g, respectively. PW/PE-OBC chips show the similar phase change temperatures of PW but with the lower latent heat about 57 J/g attributed to the less than one-third PW content in the chips. Given that the prepared PW/PE-OBC chips have the suitable phase transition temperature scope for human body and high latent heat for heat storage, it has the potential to be used for manufacture of TRFs in melt spinning process.

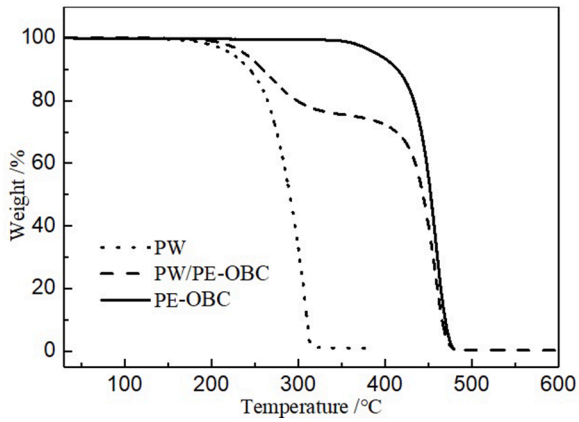


Fig. 3. The thermal decomposition curves of PW, PW/PE-OBC and PE-OBC.

Besides, the thermostability of PW/PE-OBC chips was characterized by TG. As shown in Fig. 3, one can see that compared with a single weightless stage of PW and PE-OBC occurred in the temperature ranges of 150-320 °C and 350-500 °C, respectively, there are two weightless stages on the curves of PW/PE-OBC chips, in which the first weight loss stage consistent with that of PW within 200~320 °C has 30 % weight loss as same as the feeding ratio of PW and another stage in the range of 350~500 °C belongs to the decomposition of the mixed PE and OBC. It is found that the initial volatilization temperatures of PW and PW/PE-OBC are at 195.43 °C and 224.91 °C, respectively, which implies that the restriction of PW molecules in the network structure of PE-OBC effectively inhibits the volatilization of PW and improves the thermal stability of PW/PE-OBC chips. It is concluded that the high volatilization temperature of PW/PE-OBC at 224 °C further confirms its promising application in melt spinning process of PP.

To further illustrate the different effects of PE-OBC and PE on PCM, Fig. 4 shows the different skeletal structure of PE and PE-OBC. It is seen that OBC can closely wrap PW blocks to improve the network structure of PE-OBC substrate due to its excellent elasticity and toughness. The soft elastic PE-OBC skeletal structure can clearly seal PW better than the rigid PE network, and endows PW higher thermostability.

3.2. Thermal energy storage properties of the TRFs

In practice of the melt spinning, the TRFs with PW/PE-OBC as the core and PP as the sheath were smoothly manufactured in the sheath-core ratios of 7:3, 6:4 and 5:5 on a bi-component industrial machine, and the related DSC curves and data are shown in Fig. 5.

As shown in Fig. 5, during the heating and cooling process, all of the TRFs show two peaks of melting and crystallization as same as those of PW in Fig. 2. In Table 3, it is clear that the transition temperatures of the TRFs in melting and crystallization are around 30 °C and 22°C, respectively, which are comfortable temperature ranges for the human body. And with the increasing contents of PP sheath the solid-liquid

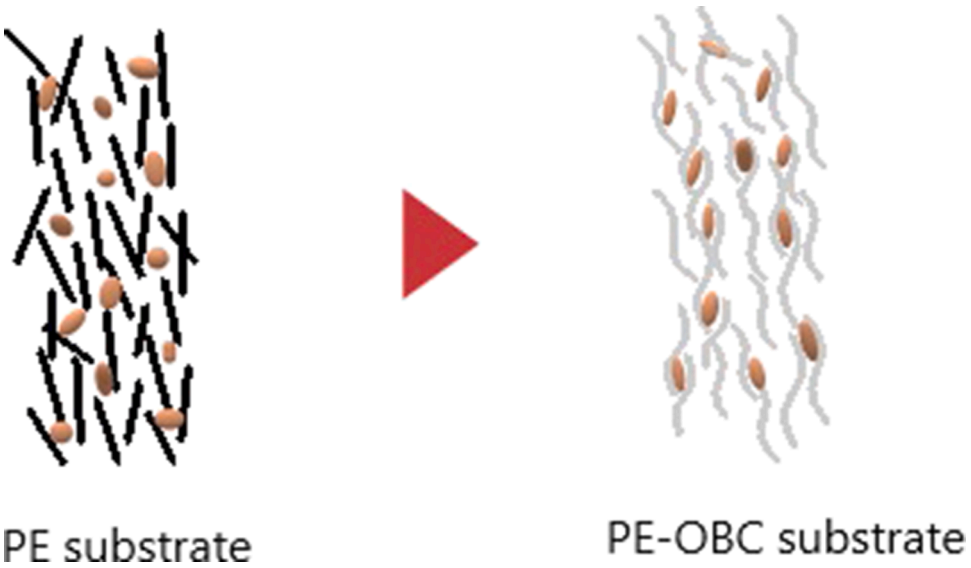


Fig. 4. The different skeletal structure of PE and PE-OBC.

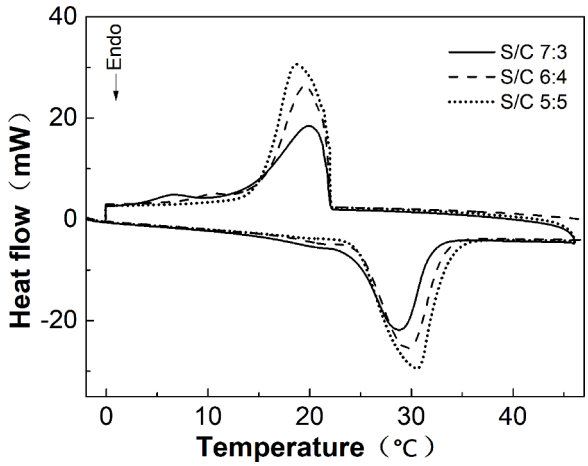


Fig. 5. DSC curves of the TRFs with different sheath-core ratio at 7:3, 6:4 and 5:5.

Table 3
DSC data of the TRFs in different sheath-core ratios.

Sheath-core ratio	Heating process		Cooling process	
	T_m (°C)	ΔH_m (J/g)	T_c (°C)	ΔH_c (J/g)
7:3	30.21	16.85	22.38	15.80
6:4	29.32	23.05	22.92	22.98
5:5	29.01	26.96	23.40	26.39

phase transition temperatures of the TRFs become slightly delayed mainly due to the heat resistance effect of PP shell during the DSC measurements. Moreover, it is found that $\triangle H_m$ of all the TRFs is very close to $\triangle H_c$ in heating-cooling cycles, indicating that the TRFs keep a good reversible phase change process as the PW as well. With the increasing contents of the core material, $\triangle H_m$ of the TRFs gradually increases from 16.85 J/g of sheath-core 7:3 to 26.96 J/g of sheath-core 5:5, being very close to the theoretical value calculated from the content of PCM in the fibers. In brief, compared with those reported data of other TRFs [23,28], the prepared TRFs in our study have higher latent heat and more preferable phase change temperature range, which seems to be promising for the temperature regulating textiles.

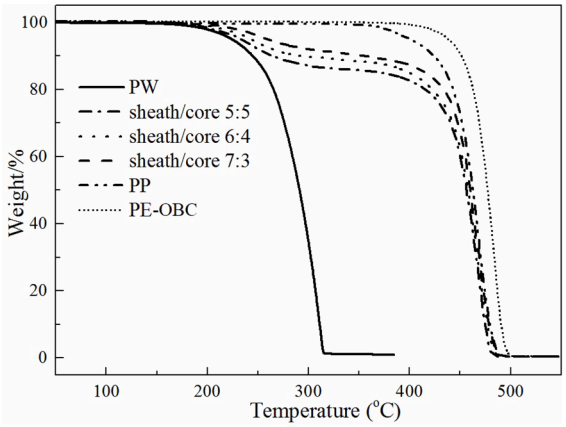


Fig. 6. The thermal weight loss curves of PW, PP, PE-OBC and TRFs in different sheath-core ratios.

3.3. Thermal stability of the TRFs

Fig. 6 shows the thermal weight loss curves of PW, PP, PE-OBC and TRFs, In the temperature range of 200-350 °C, PW has a clear volatilization process accompanied by a significant weight loss. Different from PW, the TRFs exhibit two weightless stages. The initial volatilization temperatures of the TRFs are recorded at 238.19 °C, 242.93 °C and 245.91 °C corresponding to sheath-core ratios of 5:5, 6:4 and 7:3, which have a large increment compared with PW and PW/PE-OBC chips, suggesting the better thermostability. And the mass percent of PW in the TRFs during the first weight loss stage were calculated to be 13.89%, 10.93% and 8.41% with the sheath-core ratios of 5:5, 6:4 and 7:3, respectively, which are slightly lower than the theoretical values mainly due to the volatilization loss of PW at the high melting spinning temperature.

Normally, the PCM inside the TRFs would gradually lessen following lots of recycling. Fig. 7 shows a series of DSC curves of the composite fiber with sheath-core ratio of 6:4 in the range of 0-50°C for 50 cycles. It can be clearly seen the DSC curves are almost consistent except a little decrease of phase transition enthalpy, indicating that the temperature regulating performance of the composite fiber is almost unaffected during repeated heating and cooling cycles, which confirms a good thermal cycle stability of the TRFs.

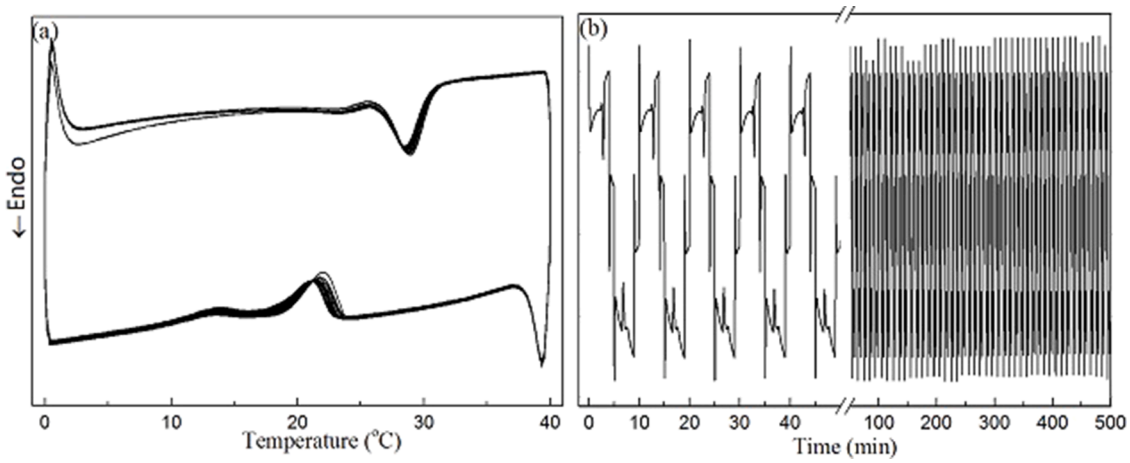


Fig. 7. The DSC recycling curves of the TRFs with sheath-core ratio of 6:4 (a) varying with temperature; (b) varying with time.

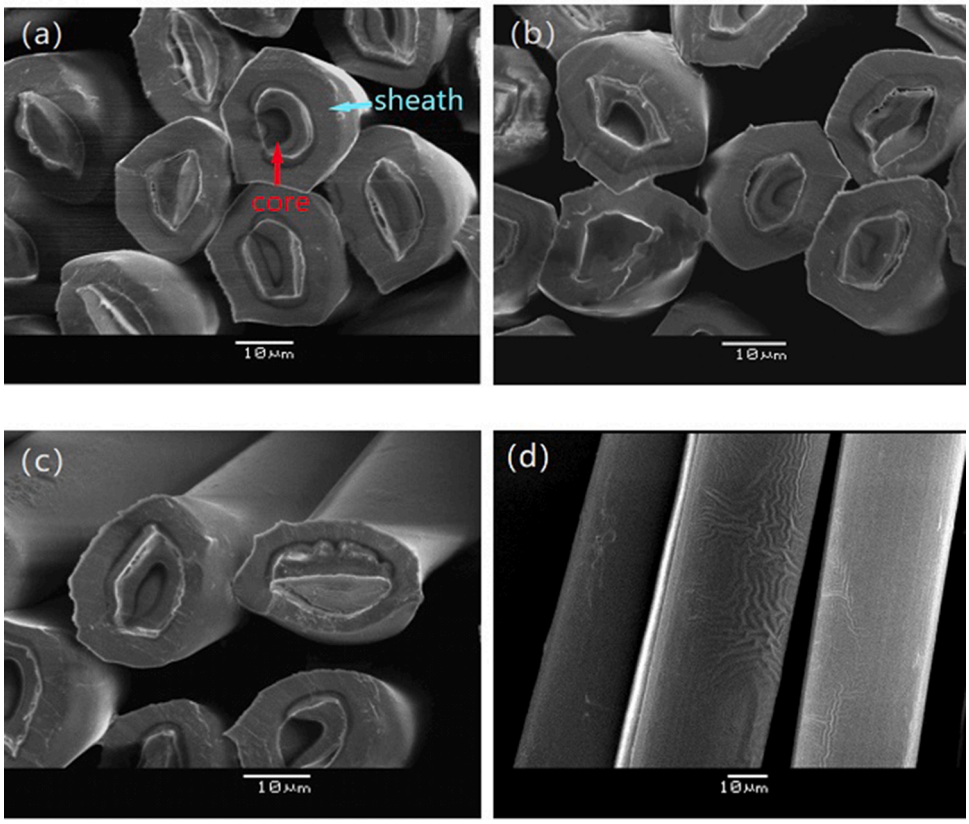


Fig. 8. SEM images of the TRFs in different sheath/core ratios (a) 7: 3; (b) 6: 4; (c) and (d) 5: 5.

3.4. Morphological structure

Fig. 8 shows the cross section morphology of the TRFs prepared in different sheath/core ratios. In Fig. 8(a)–(c), the TRFs show a distinct sheath-core structure, and there is a clear interface between the core and the sheath, which indicates that PP and PW/PE-OBC are separated from each other because of the different shrinkage rates in spinning and drawing processes. In addition, it can be clearly seen that the core layer is well incased in the sheath, and the sheath gradually becomes thinner with the decrease of the sheath-core ratios. And the obvious dimples on the cross section of the core layer are mainly due to the volatilization of some PW during the specimen preparation of SEM.

As shown in Fig. 8 (d), the TRFs have smooth and flat longitudinal surface, which indicates the addition of PW/PE-OBC as the core material had little effect on the integrity of the sheath. In our study, the temperature profile is set at 160~230 °C in the core component screw zone and 220 °C in the spinneret region, which ensures the little volatilization of PW during the whole spinning process including screw zone and spinneret region, and contributes to the good mechanical properties of TRFs due to the perfect and undamaged sheath-core structure of the fibers as well.

3.5. Linear density and mechanical properties

Strength and elongation are the most important mechanical performance indexes of fibers and can directly determine whether TRFs are

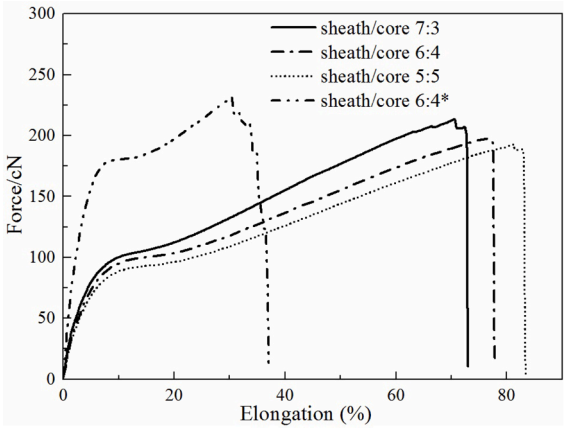


Fig. 9. The strength extension curves of the TRFs in different sheath-core ratios and drawing ratios.

Table 4
The mechanical properties of the TRFs in different sheath-core ratios.

Sheath-core ratio	Linear density (dtex)	Elongation (%)	Breaking strength (cN/dtex)	Initial modulus (cN/dtex)
7:3	169.2(3)	70.6(7)	1.70(8)	17.61(6)
6:4	168.8(5)	76.7(4)	1.53(4)	15.82(5)
5:5	169.1(2)	81.3(8)	1.41(5)	14.95(7)
6:4*	93.3(2)	30.4(3)	2.09(6)	57.27(6)

suitable for twisting and weaving processes. Fig. 9 presents the load-extension curves of the TRFs, showing the typical elastic deformation, elongation and fracture stages. Table 4 indicates the linear densities and mechanical parameters of the TRFs, in which the sample of sheath-core 6:4* represents the TRFs is further drawn at a drawing ratio of 2.0. The linear densities of another three TRFs drawn at 700 m/min are all around 169 dtex due to the similar density of the components and same spinning conditions, nevertheless, for the TRFs of sheath-core 6:4*, its linear density decreases to 93 dtex after further drawing at a winding speed of 1400 m/min.

From Table 4, the tensile strength is a function of the core material contents of the three samples drawn at 700 m/min. With the increase of the core contents in the TRFs, the breaking strength decrease from 1.70 to 1.41 cN/dtex and the initial modulus lowers from 17.61 to 14.95 cN/dtex, but the elongation of the TRFs increases from 70.6% to 81.3%. For the TRFs of sheath-core 6:4*, the breaking strength increases to 2.09 cN/dtex and elongation decreases to 30.4%. As is well known, drawing can promote the further crystallization and orientation of the polymers in fibers, which might improve the strength of the fiber and decrease the elongation to some degree. To sum up, it is thought that the TRFs have the great potential in textile industrial production as far as the thermal storage and mechanical properties are concerned.

3.6. *Weaving and thermal properties of the temperature-regulating fabrics*

Owning to the good mechanical performance and suitable linear density, as shown in Fig. 10, the TRFs of sheath-core 6:4* were successfully blended and twisted with cotton fibers in the yarns with TRF/Cotton ratios of 5:5 and 7:3, and then woven into the plain fabrics in the structure parameters of R_j=R_w=2, S_j=S_w=1, No. 50 reed and 220 × 260 (root /10 cm) on a shuttle loom. As a comparison, a pure cotton fabric was also weaved in accordance with the above parameters.

Fig. 11 (a) and Table 5 present the phase change temperatures and the latent heat of the woven fabrics characterized by DSC. It is found that the TRF/cotton woven fabrics have the similar melting and crystallization processes as PW in the range of 21- 31 °C, and show a good heat storage performance with the latent heat of 11.40 J/g for TRF/cotton 5:5 and 15.23 J/g for TRF/cotton 7:3, respectively. Compared to cotton fabric, the temperature regulation differences ΔT_d were also estimated from Fig. 11 (b) and listed in Table 5.

Fig. 11(b) shows the thermal delay curves of the TRF/cotton woven fabrics when the fabric samples are cooled down from 45 °C to 10 °C. Compared to the pure cotton sample, both of the TRF/cotton woven fabric samples present the significant heat preservation stages, starting from 28 °C to 18 °C and keeps for about 600 s, and show the temperature regulation differences of 5 °C and 3 °C in accordance with their latent heat values which implies that the thermal delay ability of the TRF/cotton fabrics is determined by their heat storage performances.

In summary, the TRF/cotton woven fabrics have a definite thermo-regulatory capacity due to the addition of TRFs, which can buffer the temperature fluctuation to some extent by thermal recycling under the heat transfer of the external energy such as solar or human body radiation.

4. Conclusion

In this study, the thermal-storage and temperature-regulated composite fibers (TRFs) are successfully prepared with PP as the sheath and PW/PE-OBC as the core by the sheath-core melting spinning process. TRFs have the apparent phase change in the temperature range of 21~30°C with a latent heat up to 26.96 J/g and present good thermal cycling performance.

After drawn at a winding speed of 1400 m/min, TRFs prepared in

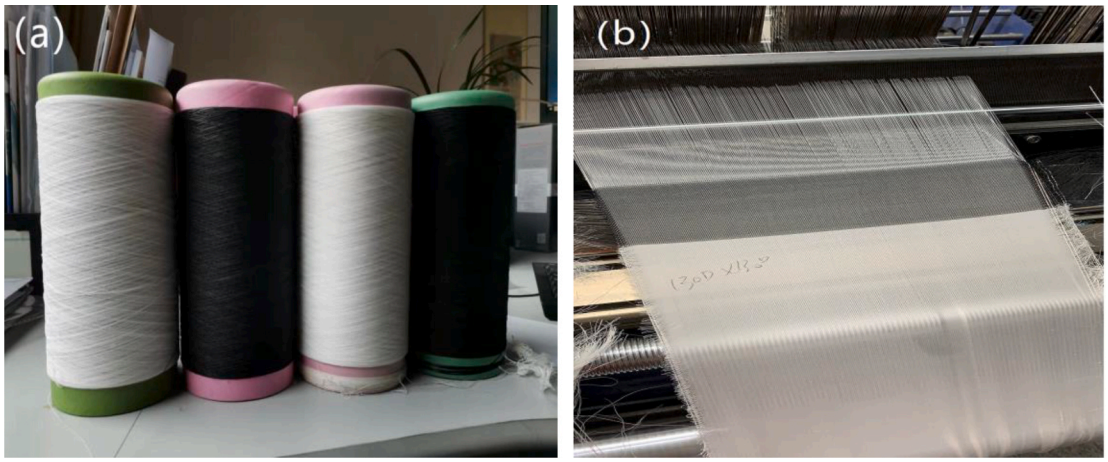


Fig. 10. The TRF/cotton yarns (a) and the TRF/cotton woven fabrics (b).

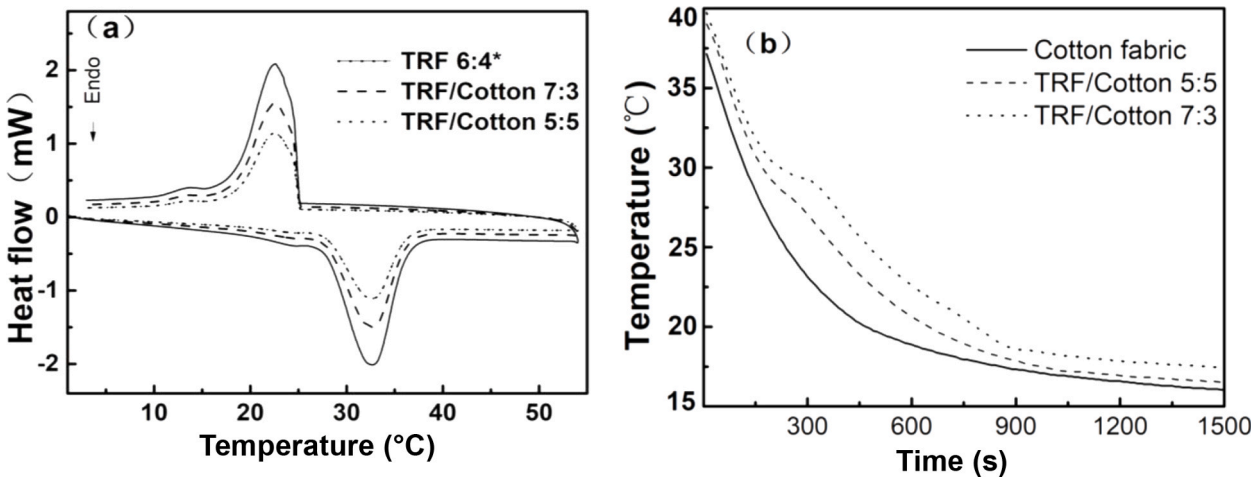


Fig. 11. The DSC (a) and the thermal delay (b) curves of the cotton and TRF/cotton woven fabrics.

Table 5
The thermal properties of the TRF/cotton woven fabrics.

Samples	TRF/ cotton	T_m (°C)	ΔH_m (J/ g)	T_c (°C)	ΔH_c (J/ g)	ΔT_d (°C)
1	0:1	-	-	-	-	-
2	5:5	31.73	11.40	21.23	11.67	3
3	7:3	32.17	15.23	21.60	16.04	5

sheath-core ratio of 6:4 show a breaking strength of 2.09 cN/dtex and an elongation of 30.4%, meeting the requirements of weaving processing. Then the temperature regulating fabrics are woven by using the prepared TRFs and cotton and tested the thermal preservation effect. The results display a definite thermoregulatory capacity during temperature fluctuation.

It is believed that the above research provides the valuable preparation methods and experimental data in the large-scale industrial production and application of the intelligent temperature-adjustable fibers.

CRediT authorship contribution statement

Guoqing Zhang: Supervision, Conceptualization, Writing – original draft. **Shixiang Xu:** Methodology. **Mengyu Du:** Investigation. **Guojin Liu:** Data curation. **Lan Zhou:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The work was supported by the National Natural Science Foundation of China (Grants No. 22075252) and the Zhejiang-Czech Industrial R&D Project (Project No.:2019C54003) .

References

[1] Z. Zhang, X. Zhang, H. Shi, W. Li, J. Meng, Thermo-regulated sheath/core submicron fiber with poly(diethylene glycol hexadecyl ether acrylate) as a core, Text. Res. J. 86 (2016) 493–501.
[2] X. Huang, C. Zhu, Y. Lin, G. Fang, Thermal properties and applications of microencapsulated PCM for thermal energy storage: a review, Appl. Therm. Eng. 147 (2019) 841–855.
[3] A. Nejman, M. Cieślak, The impact of the heating/cooling rate on the thermoregulating properties of textile materials modified with PCM microcapsules, Appl. Therm. Eng. 127 (2017) 212–223.
[4] H. Hong, Y. Pan, H. Sun, Z. Zhu, C. Ma, B. Wang, W. Liang, B. Yang, A. Li, Superwetting polypropylene aerogel supported form-stable phase change materials with extremely high organics loading and enhanced thermal conductivity, Sol. Energy Mater. Sol. Cells 174 (2018) 307–313.
[5] X. Huang, G. Alva, Y. Jia, G. Fang, Morphological characterization and applications of phase change materials in thermal energy storage: a review, Renew. Sustain. Energy Rev. 72 (2017) 128–145.
[6] H.T. Xu, N. Wang, C.Y. Zhang, Z.G. Qu, F. Karimi, Energy conversion performance of a PV/T-PCM system under different thermal regulation strategies, Energy Convers. Manage. 229 (2021), 113660.
[7] A. Karaipekli, A. Bicer, A. Sari, V.V. Tyagi, Thermal characteristics of expanded perlite/paraffin composite phase change material with enhanced thermal conductivity using carbon nanotubes, Energy Convers. Manage. 134 (2017) 373–381.
[8] L. Peng, W. Fan, D. Li, S. Wang, Z. Liu, A. Yu, X. Jiang, Smart thermal management textiles with anisotropic and thermoresponsive electrical conductivity, Adv. Mater. Technol. 5 (2020), 1900599.
[9] P. Yucan, C. Yi, Advanced textiles for personal thermal management and energy, Joule 4 (2020) 724–742.
[10] M.M. Farid, A.M. Khudhair, S.A.K. Razack, et al., A review on phase change energy storage: materials and applications, Energy Convers. Manage. 45 (2004) 1597–1615.
[11] K. Murat, K. Mahkamov, Solar energy storage using phase change materials, Renew. Sustain. Energy Rev. 11 (2007) 1913–1965.
[12] P. Sánchez, M.V. Sánchez-Fernandez, A. Romero, et al., Development of thermo-regulating textiles using paraffin wax microcapsules, Thermochim. Acta 498 (2010) 16–21.
[13] K. Iqbal, D. Sun, Development of thermo-regulating polypropylene fibre containing microencapsulated phase change materials, Renew. Energy 71 (2014) 473–479.
[14] E. Pakdel, M. Naebe, Lu Sun, X. Wang, Advanced functional fibrous materials for enhanced thermoregulating performance, ACS Appl. Mater. Interfaces 11 (2019) 13039–13057.
[15] G.J. Chen, Y.P. Su, D.Y. Jiang, L.J. Pan, S. Li, An experimental and numerical investigation on a paraffin wax/graphene oxide/carbon nanotubes composite material for solar thermal storage applications, Appl. Energy 264 (2020), 114786.
[16] K.X. Dong, N. Sheng, D.Q. Zou, C. Wang, K. Shimono, T. Akiyama, T. Nomura, A high-thermal-conductivity, high-durability phase-change composite using a carbon fibre sheet as a supporting matrix, Appl. Energy 264 (2020), 114785.
[17] P. Zhang, Z.N. Meng, H. Zhu, Y.L. Wang, S.P. Peng, Melting heat transfer characteristics of a composite phase change material fabricated by paraffin and metal foam, Appl. Energy 185 (2017) 1971–1983.
[18] S. Liang, Q. Li, Y. Zhu, K. Chen, C. Tian, J. Wang, R. Bai, Nanoencapsulation of n-octadecane phase change material with silica shell through interfacial hydrolysis and polycondensation in miniemulsion, Energy 93 (2015) 1684–1692.
[19] Y. Konuklu, H.O. Paksoy, M. Unal, Nanoencapsulation of n-alkanes with poly (styrene-co-ethylacrylate) shells for thermal energy storage, Appl. Energy 150 (2015) 335–340.
[20] A. Sari, C. Alkan, DK. Döğüşcü, C. Kizil, Micro/nano encapsulated n-tetracosane and n-octadecane eutectic mixture with polystyrene shell for low-temperature latent heat thermal energy storage applications, Sol. Energy 115 (2015) 195–203.
[21] Y. Fang, H. Wei, X. Liang, S. Wang, X. Liu, X. Gao, Z. Zhang, Preparation and thermal performance of silica/n-tetradecane microencapsulated phase change material for cold energy storage, Energy Fuels 30 (2016) 9652–9657.
[22] K. Iqbal, A. Khan, D. Sun, M. Ashraf, A. Rehman, F. Safdar, A. Basit, H.S. Maqsood, Phase change materials, their synthesis and application in textiles-a review, J. Text. Inst. 110 (2019) 625–638.
[23] K. Iqbal, D. Sun, Development of thermal stable multifilament yarn containing micro-encapsulated phase change materials, Fibers Polym. 16 (2015) 1156–1162.
[24] X. Zhang, J. Chen, Y. Zeng, Morphology development of helical structure in bicomponent fibers during spinning process, Polymer 201 (2020), 122609.

[25] J. Wu, R. Hu, S. Zeng, W. Xi, G. Tao, Flexible and robust biomaterial microstructured colored textiles for personal thermoregulation, ACS Appl. Mater. Interfaces 12 (2020) 19015–19022.
[26] X. Zhang, X. Wang, L. Hu, J. Niu, R. Yin, Spinning and properties of PP/PEG composite fibers for heat storaging and thermoregulated, J. Tianjin Inst. Text. Sci. Technol. 1 (1999) 3–6.
[27] X.X. Zhang, X.C. Wang, H. Zhang, J.J. Niu, R.B. Yin, Effect of phase change material content on properties of heat-storage and thermo-regulated fibres nonwoven, Indian J. Fibre Text. Res. 28 (2003) 265–269.
[28] C. Cherif, N.H. Tran, M. Kirsten, H. Bruenig, R. Vogel, Environmentally friendly and highly productive bi-component melt spinning of thermoregulated smart polymer fibres with high latent heat capacity, Exp. Polym. Lett. 12 (2018) 203–214.
[29] M.M. Umair, Y. Zhang, K. Iqbal, S. Zhang, B. Tang, Novel strategies and supporting materials applied to shape-stabilize organic phase change materials for thermal energy storage-a review, Appl. Energy 235 (2018) 846–873.
[30] Y. Tang, Y. Jia, G. Alva, X. Huang, G. Fang, Synthesis, characterization and properties of palmitic acid/high density polyethylene/graphene nanoplatelets composites as form-stable phase change materials, Sol. Energy Mater. Sol. Cells 155 (2016) 421–429.
[31] A. Trigui, M. Karkri, I. Krupa, Thermal conductivity and latent heat thermal energy storage properties of LDPE/wax as a shape-stabilized composite phase change material, Energy Convers. Manage. 77 (2014) 586–596.
[32] J.-.L. Zeng, F.-.R. Zhu, S.-.B. Yu, Z.-.L. Xiao, W.-.P. Yan, S.-.H. Zheng, et al., Myristic acid/polyaniline composites as form stable phase change materials for thermal energy storage, Sol. Energy Mater. Sol. Cells 114 (2013) 136–140.
[33] D.G. Atinafu, W. Dong, C. Wang, G. Wang, Synthesis of porous carbon from cotton using an $Mg(OH)_2$ template for form-stabilized phase change materials with high encapsulation capacity, transition enthalpy and reliability, J. Mater. Chem. A 6 (2018) 8969–8977.
[34] M. Karthik, A. Faik, P. Blanco-Rodríguez, J. Rodríguez-Aseguinolaza, B. D'Aguanno, Preparation of erythritol–graphite foam phase change composite with enhanced thermal conductivity for thermal energy storage applications, Carbon 94 (2015) 266–276.
[35] J.-.L. Zeng, J. Gan, F.-.R. Zhu, S.-.B. Yu, Z.-.L. Xiao, W.-.P. Yan, et al., Tetradecanol/expanded graphite composite form-stable phase change material for thermal energy storage, Sol. Energy Mater. Sol. Cells 127 (2014) 122–128.
[36] B. Mu, M. Li, Fabrication and thermal properties of tetradecanol/graphene aerogel form-stable composite phase change materials, Sci. Rep. 8 (2018) 8878.
[37] Z. Xiangfa, X. Hanning, F. Jian, Z. Changrui, J. Yonggang, Preparation, properties and thermal control applications of silica aerogel infiltrated with solid–liquid phase change materials, J. Exp. Nanosci. 7 (2012) 17–26.
[38] X. Huang, G. Alva, L. Liu, G. Fang, Microstructure and thermal properties of cetyl alcohol/high density polyethylene composite phase change materials with carbon fiber as shape-stabilized thermal storage materials, Appl. Energy 200 (2017) 19–27.